

Climate Change for Astronomers

Causes, consequences, and communication

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Chapter 19

The Future of Meetings

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“In our view of the future, it is not so much distance that will be abolished, but rather our current concept of being there.”

—Jim Hollan & Scott Stornetta “*Beyond Being There*” (CHI, 1992)

Astronomy has a long history of innovative global collaboration, stretching back more than a century. Over the course of the last few decades, the discipline has embraced many technologies to better facilitate the dissemination of information. Yet the way in which we actually physically meet to partake in this dissemination has remained relatively fixed in format. During the COVID-19 pandemic we saw how our community could convene productively online. While virtual conferencing is very much in the “innovation stage” we also saw many immediate benefits, not only from drastically reduced carbon emissions but also by making it possible for many to participate who were previously excluded. With the rising impacts of climate change it is imperative—now more than ever—that we look to new ways of collaboration to improve the long-term impacts of our meetings on both our discipline and our planet. In this chapter, The Future of Meetings (TFOM) community presents some of the lessons we have learned in our quest for a “new normal” for academic meetings. We discuss some of the common challenges around online and hybrid interactions, and solutions to them. We conclude by looking to the future, whereby technology has the potential to dramatically improve how we work, collaborate, and socialize in an increasingly connected world.

19.1 Introduction

A century ago, there were no commercial flights, no computers, no internet and no smartphones. The world was far less connected than our contemporary one, especially when we consider the immediacy with which we can now spread information around the globe. Yet the way people back then met for conferences,

meetings and collaboration actually looked remarkably similar to today, involving individuals or groups traveling sometimes great distances at often high cost to spend a relatively short period of time together.

Given then how much has changed about the world from the 1920s to the 2020s, it is curious that astronomy, academia and even society at large still consider the week-long conference as the gold standard for facilitating effective distributed collaboration, despite the fact that we can essentially pick up an electronic device of our choosing (telephone, tablet, laptop, mixed reality headset) and meet people from anywhere on Earth, as long as there is an internet connection bridging the two devices. Because the traditional in-person meeting format has existed for so long, we assume that it must be more effective, more useful and more beneficial than any other mechanism. So it has always seemed “worth it” to traverse the globe to meet and collaborate with colleagues—because that is how things are done, at least for those who have had the means to do so.

But we have a dilemma: as a global community, we recognise the need to reduce our carbon footprints urgently, by large amounts. An individual’s carbon footprint is made up of a complex number of factors, with details outlined elsewhere in this book. But a significant component for an astronomer (and more importantly, one we have *direct control* over) is the extent to which we travel and the means by which we do it. Many discussions about sustainability in astronomy focus on all the things that might be lost when we change the way we work and collaborate, particularly when it comes to travel and conferences. There are commonly-raised objections to online and hybrid interaction, many of which have a basis in reality but the majority of which have solutions. It is not just sustainability which drives the need for change: considerations of accessibility, inclusivity and economics are major factors that encourage us to look for innovative solutions to distributed collaboration.

What we aim to show you in this chapter is that not only has astronomy always been an evolving field with a strong dependence on technological innovation for progress, but also that we gain much more than we potentially lose when we adopt enabling technologies to improve the ways in which we collaborate. It’s early days still, and it remains to be seen which options we ultimately choose as a community. But our experience thus far has shown that the doors which open in a community that embraces online means of communication, and the seats at the table offered to those who have traditionally been denied in-person opportunities, are very much worth the effort, and they point to a brighter, more inclusive and more sustainable future for astronomy.

19.2 The Evolution of Meetings and Collaboration in Astronomy

Astronomy has a history of innovative global collaboration that goes back more than a century. This has been a natural requirement for a field under one sky that needs to share data from all parts of the world, crossing not only international lines but political and social barriers. Scientific conferences, in which astronomers from all over the world meet, have played a key role in the development of these collaborations as well as the agreements on conventions needed to work



Figure 19.1. Attendees of the first IAU assembly in 1922, in Italy. Credit: IAU.

harmoniously together. The International Astronomical Union (IAU) was formed “to facilitate the relations between astronomers of different countries where international collaboration is necessary or useful.” It had its first meeting in Rome in 1922 (Figure 19.1). These conferences were also social events; and the interactions extended well beyond the meetings themselves, with participants often sharing travel by boat or train. There is almost no modern equivalent to the development of a collaboration while traveling together to distant lands.

In the following sections we look at the evolution of different aspects of astronomy conferences as examples of how changes have occurred in the past and as potential indicators of the changes we might expect in the future.

19.2.1 Poster Sessions

Poster sessions were introduced in the 1970s to solve the problem of too many participants needing to make oral presentations to get travel support. The excess of speaking participants was reducing the time available for discussion, and increasing the need for parallel sessions. Although initially disfavoured as lower status, posters slowly evolved to become an effective modern presentation platform, featuring “*sparkler talks*,” direct contact with presenters, and the possibility for more complex presentations such as interactive computer models. Poster sessions can be readily translated into an online activity, with embedded links to powerful graphics and interactive displays. Proximity social media platforms such as Gather can facilitate direct interactions. Unfortunately these do not combine well with in-person interactions in a hybrid format, but online posters could effectively replace any need for in-person interactions and printed poster displays. Indeed, virtual exhibits

and posters will probably become one of the first effective uses of virtual reality in academia.

19.2.2 Discussion Sessions

Questions and Answer (Q&A) sessions and discussions after talks are the essential difference between a meeting and a published paper. But there have been large and mostly negative changes since the introduction of digital technology. Up to about 1980, all discussions after presentations were recorded and published in conference proceedings. Meticulous processes were used to collect and verify the Q&A. This made it possible for those who couldn't attend to at least be aware of the content of the discussions, if not participate. It also provided material for historians seeking to understand the context of the scientific results being presented. This context is rarely captured in scientific publications. We have now lost some of this history because it was incorrectly assumed that new communication and recording technology would suffice. Any audio recorded at the conference is rarely transcribed accurately, the identity of the questioner is often not indicated, replaying recorded meetings is extremely time consuming, and Q&A discussions may not have been recorded or were lost. It is clear that modern technology offers great opportunities to retain the details of discussion sessions (e.g. with accurate automated transcription and AI-generated highlight summaries). But this requires active consideration and experimentation on the part of meeting organizers, as well as better standardized policies around the archiving of legacy-value meeting content.

19.2.3 Meeting Presentation Technology

Changing technology has had a big impact on how conference presentations are made, and we should expect this trend to continue. Few of the past changes could have been predicted and there is no reason to expect this will be different in future, but we can still look at past changes for examples of what can happen:

- 1960: Slides (3x4") glass—high quality but cumbersome.
- 1970-90: 35mm slides. Maintained high quality, enabled multiple projectors and remote control
- 1960-90: overhead projectors and transparent sheets—flexible and could be modified in real-time. At first these were hand drawn, then copiers were used, followed by computer generated transparencies. Some loss of quality but enabled real-time changes and more artistic personalized presentations.
- 1995: computer generated presentations using an LCD panel on overhead projectors. This was the beginning of computer generated graphics and it was to lay the path for future on-line presentations, although this was not the motivation at the time. Computer generated displays removed discrimination against presenters with poor writing skills.
- Mid 1990s: video presentations into multiple meeting spaces or lecture theaters.

- 2000: digital projectors. Initially inferior quality and brightness but still used because of convenience. Eventually they matched 35 mm slide quality. The introduction of digital projectors was very disruptive due to technical issues and immature support software, somewhat like the current technology for Virtual Reality.
- 2010: Digital projection technology finally stabilized and their use became ubiquitous.
- 2010: Video conferencing with individual participation using smartphone technology.
- 2020: COVID-19 rapidly accelerates the development and adoption of video conferencing tools. This was of course an unpredicted outcome.

However, until very recently technology has had little impact on the perceived need to meet in-person.

19.3 Why Do We Travel in Astronomy?

Digital technology is changing our research practices and society at large incredibly rapidly, but the uptake is not uniformly distributed. There also remains a widespread cultural impression that the most effective collaboration and learning occurs in person. In fact, we have built an academic system in which travel is widely accepted as necessary for career advancement. This increases astronomy's environmental impact significantly, and is highly exclusive given the many barriers to travel. Technological evolution offers a chance to improve our practices, reflect on the suitability of existing methods and shape the design of future interactions.

The need for effective communication with a global cohort seems like an ideal problem to solve with technology. Indeed, we have seen asynchronous communication platforms such as Slack become ubiquitous. However, the continuously connected nature of modern life can demand more of our attention and cause excessive context-switching. Instead of being the only way to collaborate, in-person meetings are now seen as rare opportunities to focus on a specific topic or task, with everyone in the same time zone. Here, we examine some of the most common contemporary forms of collaborative work with the intent to identify key factors.

19.3.1 Conferences and Workshops

While refereed journal publications remain the primary means of exchanging information with other astronomers, the sense of finality associated with publishing a result creates a need for other forums in which ideas can be exchanged less formally, particularly prior to publication. This role has typically been fulfilled by conferences of various structures and sizes, ranging from small, local workshops on specific topics, to large, international meetings covering the whole of astronomy and astrophysics. It has been argued that these meetings are necessary to spur ideas for new science and collaborations, by enabling people with shared research interests to meet.

The modern conference exhibits a poorly concealed duality in which its primary format (dominated by talks and other largely passive delivery of content) is often considered a prelude to the more free-form and productive discussions that take place in social settings around the periphery. Attendees often justify the use of institutional funds for travel and registration with the pretext of giving a talk or poster to disseminate their work. Yet in reality, it is this peripheral social element, the promise of meeting collaborators new and old, that is the strongest motivation for many to attend. Talks serve as an abbreviated form of introduction in which someone's current research interests are advertised to the assembled community. However, the growth of the astronomical community and proliferation of research topics has led to larger conferences needing multiple parallel sessions, such that no attendee can experience more than a small fraction of the formal content. Ironically, scientists who share research interests may be in attendance at the same conference but not meet simply because they chose to attend different parallel sessions. Some large conferences, winter meetings of the AAS for example, have upwards of thirty parallel streams. It is clear from this alone that the purpose and format of in-person meetings needs to be reassessed.

19.3.2 Education and Training

Education and training are also evolving with technology and many universities now offer online units. However, teaching complicated concepts at the postgraduate level is often done at schools that resemble conferences, with lectures on specific topics instead of research talks. Under ideal circumstances, in-person training has the advantage of rich feedback that can lead to a quicker or better understanding of the topic. However, in practice many training workshops inevitably involve periods of passive content-delivery which, as universities have shown, doesn't require face-to-face delivery to be effective in its purpose. Additionally, there is an increasing need for on-demand training that can be accessed when needed, rather than intermittent events. Finally as cohort sizes increase, physical limitations of scale become a problem. Technology can provide the tools to create effective online training material, circumventing these issues, but there are initial investment and re-training barriers to the development of quality content.

Community support is also a significant part of an astronomer's informal lifelong training, which naturally evolves alongside our social structures. It is now much easier to be connected to a larger community and therefore a wider pool of knowledge, but we still need to provide supportive forums in which questions are encouraged. How we best adapt the education and training of the next generation of astronomers based on their evolving needs is very much a work in progress across the field.

19.3.3 Observatory Visits

Technology has had the greatest impact on how astronomers build observatories and conduct our research. While it is important to support a wide range of facilities with varied roles, the largest and most data-intensive facilities tend to dominate the

research landscape and reach the largest number of astronomers. As the complexity, scale and physical isolation of our observatories increases, it becomes less feasible to directly interact with them. In fact, many observatories even of small and medium scale have transitioned to more remote operations for reasons of efficiency and cost among others. For space-based observatories, observations are of course conducted remotely, but this is an increasingly standard observation mode for ground-based telescopes as well. Indeed, many astronomers now do their research entirely using online archives that make old and new data much more accessible, increasing the data's value and therefore observatory efficiency.

It is worth noting that the transition from “classical” observing—where the astronomer is physically present at the telescope at the time observations are completed—to remote, queue and robotic observations was also initially met with resistance; e.g., when Gemini Observatory was under construction in the 1990s and the idea of queue observing was proposed many astronomers felt that it was important to be at the telescope so as to be able to make quick decisions about the observing programme based upon weather conditions. As a concession Gemini did at first offer a classical observing mode, but it soon became clear that queue observing made the observatory more efficient and enabled kinds of science not possible under classical scheduling. Most astronomers eventually accepted queue observing, recognizing that it increased scientific productivity. Many modern telescopes, such as the CSIRO ASKAP radio telescope and the Vera C. Rubin observatory, are operated such that observations, data calibration and initial analysis are fully automated.

It is of course important for astronomers to understand how data is gathered so they can identify systematic errors and extract the best possible result. Previously, this information may have been gained by visiting an observatory or control center. With increasing decentralization of observatory operations, it is important for astronomers to have clear lines of communication with the observatories they use, but the format of these interactions must also evolve. From the educational perspective, it is also important and valuable to provide opportunities for the next generation of astronomers to interact meaningfully with observatories and their staff, and it is likely that at least some of these interactions will benefit strongly from taking place in person.

19.3.4 Collaboration Trips and Job Tours

For better or worse, it is widely acknowledged that building a successful career in astronomy requires developing a network of global contacts. There are usually several research groups from different parts of the world working on similar topics; and these groups may have expertise in different aspects such as theory, simulations or observations. It is common for an early career scientist to tour a variety of research groups in an effort to understand the totality of their field, build their professional network and create future employment options. Similarly, it is common for an established scientist to take a sabbatical, dedicating several months to

pursuing a research goal in a location other than their main institution while setting other duties aside. Some aspects of these career-oriented trips are difficult to reproduce any other way, such as immersing oneself in the culture and surroundings of a potential future workplace. However with increasing freedom to work from anywhere, the need for such trips may diminish over time. Instead, we will likely need to develop a culture that supports setting aside normal duties to change focus without the need for physical separation from one's home institution.

19.3.5 Review Panels and Committees

For senior astronomers, it is typical to serve on various review panels, steering committees, resource allocation committees, and other esteemed assemblies. It has been common for such groups to meet in person, often in conjunction with a physical tour of a facility and social events. Such groups are responsible for making key decisions that impact the future of astronomy, so it is widely accepted that a significant devotion of time and travel is justified. However, since it is desirable for these panels to have diverse representation from around the world, senior astronomers tend to be responsible for disproportionately large carbon emissions. The goal of diverse representation would in fact be better served by relaxing the burden of travel and time required to participate in leadership activities. Technology can be used to create realistic virtual representations of facilities such as telescopes, virtual meeting spaces and easier access to ancillary information such as the technical feasibility of an observing proposal. Technology can also help increase the transparency of decision-making processes and organize input from a much wider range of people. Already virtual meetings are being used for review panels by many observatories and funding organizations; e.g., NASA and the NSF.

19.3.6 Building and Fortifying Distributed Communities

The experimental nature of research means that learning the process is just as important as understanding the background. While it is possible to teach the principles of the scientific method, knowing how to apply these principles has typically required hands-on experience. Knowledge is handed on from experienced astronomers to their postdocs and students. Research groups form as semi-autonomous units, then collaborations form between these groups. Typically, individual research groups have been co-located to facilitate rapid communication. However, the increasing scale of astronomy has led to groups and collaborations numbering hundreds of researchers, which means it is no longer feasible for everyone to be co-located. It is also increasingly difficult to know the skills and talents of everyone in the group, and yet without these sprawling groups with diverse skill sets it would not be possible to operate complex projects such as the Event Horizon Telescope and the LSST survey. Local interactions are still important, but to be an effective part of the global community, we must constantly seek to improve our ability to collaborate across large distances.

19.4 The Cost of “Business as Usual” Astronomy Practices

There will remain in the astronomy community an existing need for travel to some extent for the reasons listed above, but at what cost? It is time to rethink the continuation of these activities in person, as the disadvantages of “business as usual” far outweigh the benefits of alternative approaches. There is no doubt about the carbon footprint of travel. However, it has only recently been recognised how the disadvantaged are further marginalized by the way we do things now. We need to address problems with accessibility and inclusivity so the whole community can participate on a level playing field.

19.4.1 Sustainability

We live on a planet with finite resources, and in the current era, our travel is powered almost exclusively by fossil fuels. The fact that burning fossil fuels directly causes climate change cannot be ignored. Anthropogenic climate change is everybody’s problem, but the responsibility to mitigate it lies particularly with those with the power, influence and ability to change our practices. As discussed in Chapter 18, travel to an in-person meeting averages 1–3 tons CO₂e per participant per meeting, with variation depending primarily on the distance traveled (Burtscher et al. 2020, Klöwer et al. 2022, Gokus et al. 2023). For perspective this can be more than an average individual in a developing country produces *per year* (IEA 2023). In order to reach the Paris Target limiting global warming to 1.5°C above the pre-industrial average, the global carbon budget needs to be just 2.3 tons of CO₂e per person per year by 2030 (Carbon Independent 2007). It is impossible to achieve this target while undertaking regular long-distance travel, and indeed most people with access to higher education enjoy lifestyles with carbon budgets many times this goal. As such, we have an urgent need to reduce emissions for the sake of future generations to have the opportunity to continue to engage in science and astronomy, in a world that remains habitable to humans. If the reasons we travel, as discussed above, can be at least partially replaced by online meetings and collaboration, then every bit of carbon not emitted into the atmosphere is doing good for the planet and all who live on it.

19.4.2 Accessibility and Inclusivity

There are many reasons to adopt virtual interaction beyond addressing climate change. In particular we need to acknowledge that in-person meetings as they are done now are exclusive in part because they are inaccessible to many. The term “accessibility” is frequently associated with the physically disabled, but there are more factors that make in-person conferences out of reach for many people; e.g., financial, geopolitical, family and job responsibilities. International travel in particular is skewed towards the privileged, those with the financial capacity to afford increasingly costly air travel, accommodation and associated expenses. Furthermore, many people are excluded through visa restrictions, bureaucracy and fees. Moving conferences online removes many of these barriers and attendance

figures for virtual meetings have shown an increased participation of a far more diverse audience (e.g., Skiles et al. 2022).

But bringing people to a conference is only half the story. Once they are there they should feel welcome, as highlighted by the associated idea of inclusivity. Conferences often have a poor reputation regarding the safety of junior researchers—especially female—in traditionally male-dominated fields such as physics (Aycock et al. 2019). Online conferences can provide a safe environment for those who feel uncomfortable in such settings, but that does not mean they are necessarily welcoming. It is a frequent criticism that online conferences lack the human touch, and they can propagate existing inequalities; e.g., women who do not travel are less likely to be relieved of obligations at home during a virtual conference, which decreases their ability to fully participate in the event (NASEM 2021). Likewise, X (formerly Twitter) is a common communication tool for conferences, but users tend to interact more with people with similar stature or those from their region, mirroring aspects of inequality inherent to in-person conferences (Duncan & Shean 2023). But with good design social interactions can be improved, as will be discussed in the next section.

19.5 Technology as a Facilitator for Interaction

In 1985, the first network connecting institutes at the US and European space agencies was created across the Atlantic. The purpose was to rapidly share mission data from the International Cometary Explorer that had to be downloaded in Baltimore, USA, but analyzed in Darmstadt, Germany. The network reduced a task to a matter of hours that had previously taken months of recording data tapes and mailing internationally, and enabled the interception of a spacecraft with a comet to be broadcast live to a public audience.

The network was later extended to Japan and Russia, revolutionizing the ease in which the world's largest space agencies could work together. Tasks that previously required multiple long-distance trips to install instruments on joint missions could now be done remotely, making the collaboration process vastly simpler.

This early internet was very limited by today's standards, but it allowed real-time communication across huge distances. This meant people could interact as if their collaborators and equipment were in a single building. By the 2010s, message exchange was being supplemented by video conferencing, allowing regular meetings to be held easily with attendees from all over the world. The advent of smart phones similarly normalized distributed communications and increased the critical mass of people worldwide who relied on technology to stay connected with distributed family, friends and colleagues.

While incredibly useful at stripping away the restrictions of distance, online communication has also become widely used for even local teams due to the ease of supporting asynchronous discussion. Video call platforms are particularly successful for seminars and meetings where there is a one-way flow of information at a single time to a large group of people, or a live discussion between a small number of individuals. But for larger teams that may also need to communicate

asynchronously over a long period of time, collaborative platforms aimed at creating online communities have grown in prominence.

Examples of this include Discord on the social front as well as Slack and Microsoft Teams for professionals, where groups of people can join a virtual space dedicated to their team, organization, or event and exchange messages, share media, and also host video meetings with other members. Slack now reports more than 10 million daily active users across more than 150 countries, Discord at 26 million active daily and 125 million a month (Curry 2023a), while Microsoft states a monthly figure of more than 270 million active users (Curry 2023b).

These kinds of spaces have proved particularly successful for group discussions, which are difficult to achieve via email as only the most recent contribution is easily seen, and also challenging for both in-person and video meetings where a single time must be agreed upon, and a small number of voices often dominate the discussion.

As a result, collaborative platforms such as these are used by organizations, local and disparate working groups, and also for conferences. The latter usage provides a place to ask questions about presented talks and posters to the speaker, and also see replies to questions from other attendees. This is useful both not only for on-line conferences, but also for in-person meetings where Q&A time is limited.

Through such advances, technology has been able to provide wide and varied ways of interacting online that come with added advantages and some pitfalls. By far the main advantage concerns accessibility and inclusivity. In addition to removing the barrier of mobility, the online environment can enhance the conference experience for people with a range of disabilities, such as through the use of closed captions for the hard of hearing and audio description for the visually impaired. Astronomy can go beyond the common senses (Vargas et al. 2022) through many accessibility tools enabling equal participation of physically and neurologically disadvantaged people (Tang 2021). Specifically for inclusivity, instant translation can break through language barriers for people from different nations.

While it should be recognised that online accessibility does not automatically translate to a welcoming experience, this is mostly an issue of implementation rather than the format itself. Too often online conferences are focused on the in-person component without taking time zones into account or giving presence to the online audience who are often treated as passive spectators. With good design, an online conference can be made to feel inclusive of the audience through structured sessions, such as “onboarding” to familiarize the participants with the online platform and tools or dedicated time to meet for discussion, while **unstructured sessions such as games and shared breaks can be used to keep the online participants connected**. The biggest issue of timezone can be mitigated by not being too reliant on synchronous content, with asynchronous viewing of recorded talks and the aforementioned tools, it is possible to keep questions and discussions going “after hours.”

The result is that online technology greatly facilitates interaction and communication, regardless of whether physical distance is an issue. For situations where

distances are great, these tools help remove the need to travel large distances and allow the easy, regular contact required to build a successful project.

Creative responses in the time of COVID leveraging technology to work more effectively online have shown that it is possible to move forward from “business as usual.” Already we are seeing changes in the way we work as many companies embrace the flexibility of working from home. Similarly, the astronomy community should be pushing for better practice for a more sustainable, accessible, inclusive future.

19.6 Where Do We Go from Here?

So given the above, what comes next? It is clear that our communication methods and approaches have evolved over time, and there remain many reasons in astronomy for meeting and interacting. What remains to be seen is how we determine when these interactions need to happen in person, given considerations of the costs and benefits, and when they can be shifted to alternative modes that are less carbon-dependent and exclusive. As a global, distributed community, we are now faced with choices each time we meet: in-person, online or hybrid? In our experience, we continue to see many questions around the feasibility of non-traditional modes, particularly when it comes to the cost of hybrid and the effectiveness of online interaction.

For context, we currently operate as a community of practice known as “[The Future of Meetings](#)” (TFOM), stemming from the 2020 symposium of the same name (as outlined below). Our community includes scientists, researchers, educators and industry representatives, all of whom are united by a common interest in how future meetings and interactions may be optimized, with a central theme of leveraging technology for improved accessibility, inclusivity and sustainability. As TFOM, we continue to experiment with and explore new approaches (e.g., [Figure 19.2](#)) and to raise awareness of the many possibilities for effective hybrid and online interaction that have not yet become standard practice within the astronomy community or elsewhere.

The pandemic offered us an extreme opportunity to reset our assumptions about meeting and proved that even with several months of not meeting in person as freely as we once did, we were able to continue working and collaborating and, in the case of astronomy, keep progressing our exploration of the universe. There were clear limitations in many cases, but often these came back to a choice of design, approach or technology rather than a fundamental limitation of online interaction.

Now, for all of the reasons outlined above, it is critical that we find a way to step back, look at the broad picture of all the ways we meet and interact, and determine an effective balance between the familiar sociability of in-person interaction and the relatively unexplored possibilities of the online format. For us in TFOM, it has been clear that the most exciting element of using technology to go online is not the endless quest to reproduce in-person interaction, but all of the wide and unexpected possibilities that arise when you embrace the concept of going “beyond being there” (Hollan & Stornetta [1992](#)).



Figure 19.2. Members of “The Future of Meetings” (TFOM) community of practice, gathered in Altspace at the conclusion of an event series run by TFOM in VR during 2022.

19.7 Addressing Common Concerns about Online and Hybrid

During the pandemic, the world experienced a near-complete shift to online formats for interaction which radically and abruptly displaced our expectations about meetings. Since mid-2021 with the return of international travel and the lessening of COVID restrictions, meetings and conferences have been evolving in many directions at once. We have seen the predominant return of the in-person only conference/meeting, as if it marks the triumph of “back to normal,” alongside a variety of types of hybrid conferences and meetings, many of which have left people concluding that hybrid interaction is ineffective, or worse, impossible. Some meetings and conferences have stayed online-only, but these are in our experience the minority compared to the other two formats. In astronomy, it seems that conferences and workshops in particular have remained resistant to the reduction of in-person reliance, whereas more functional work practices (e.g. weekly or monthly meetings, colloquia) have shifted towards online or hybrid due to the intrinsically-distributed community.

As for why online formats aren’t adopted more often as the primary format, or why hybrid is done either poorly or not at all, we have observed from the TFOM context that there are common assumptions being made across astronomy, academia, and society more broadly. In this section, we have collated a list of the most prevalent assumptions and criticisms being made of these formats, which we see as presenting barriers to the better and more widespread adoption of online formats. These barriers are not insurmountable because, in the majority of cases, they are embedded with misconceptions and there are straightforward ways to reduce the impact of these barriers.

The statements presented below were collated by the authors of this chapter, and then ordered based on the weighted consensus of their importance to include here and by how much an improvement can be made by using different methods and new tools, both of which we indicate for each statement. We aim with this section to help illuminate some of the current challenges about online and hybrid formats, and offer tangible solutions to address these for those who are organizing, attending and/or funding astronomy meetings and conferences.

19.7.1 “It’s Fine for Those with Accessibility Needs to Be Online, Everyone Else Can Go to Meetings in Person”

When it comes to hybrid formats, there has been considerable discussion about how online participants are often treated like second-class citizens, because the experience and design of most conferences is focused too much on the in-person component. Some surveys (e.g., Remmel 2021) indicate that researchers want hybrid to be the future format of meetings and conferences, but when they are asked how they personally want to attend the majority choose in-person. The issue with this is that if we continue to put most of our effort into the in-person format because that’s what we know and that’s what the majority will choose, then we will never create innovative, equitable hybrid or online-only formats. This means that those who participate online will always have a second-class experience, and miss out on opportunities that are only offered to the in-person portion of the attendees. There are many reasons why someone might choose to attend online rather than in-person, and many more reasons which remove their ability to choose freely. It is an extreme stance to deliberately relegate disadvantaged communities to online formats with little funding, planning or support, while the more privileged can continue to gather in person. This is especially true since both hybrid and online interaction can (with some thought, effort and design) be done well.

Recommendations

- Ensure every meeting and conference has reasonable (even if basic) online access.
- Give attendees, speakers and stakeholders a choice in format from the beginning.
- Host online-only gatherings where possible to give people a more equal voice.
- If a meeting must happen in person, use accessibility grants to support both in-person and online attendance, to assist diverse attendance.
- Try to establish parity. Avoid situations where only a very small fraction of attendees are online-only.

19.7.2 “You Can’t Reproduce the Organic and Serendipitous Interactions Online that You Have in Person”

The belief that people cannot have organic and serendipitous interactions online in the *same way* they have had in person may be somewhat true, but it is not true that equally effective interactions cannot be achieved online. Most online conferences

currently do not allocate time or mechanisms for the side-chats that are part of attending talks, instead focusing too much on the functional structured elements of a conference program (e.g. talks, panels). It is also important to consider the exclusivity of in-person events. While meeting someone in person may remain the gold-standard for some, think of the future collaborator *you didn't meet* because they couldn't attend.

While in-person interaction continues to hold value in the future interactions of our communities, online interaction can be just as effective and impactful when it comes to serendipitous interactions. All it relies on is for organizers to think carefully about how best to facilitate these modes of interaction, whether it is by specific organized social events or by making use of technology that helps with introductions and asynchronous interaction. In particular, immersive technology (e.g. virtual reality) presents a much more natural way to interact that is closer to our traditional expectations of body language and shared “physical” spaces.

Recommendations

- Provide specific planned mechanisms and opportunities for people to interact online.
- Host structured interactive meet-ups such as poster sessions and social events.
- Be innovative in approach and experiment with new platforms/technologies.

19.7.3 “Hybrid Is Impossible because It Doubles the Cost of the Conference”

There are a few factors mixed into this misconception that are worth unpacking. The first is that there is only one way to hold a hybrid conference, when in fact hybrid meetings range from completely passive “fly-on-the-wall” style online attendance, to an experience that is maximally equivalent whether an attendee is in-person or online. The former is far too commonly accepted as “hybrid” and the latter is still to be realized in future with the rise of widespread mixed reality technology. For now, effective hybrid formats lie somewhere in between these two extremes, and the right approach (e.g. the effort/resources invested to bridge the gap between the two audiences) should be adopted based on the situation. That said, it can be relatively cheap and low-effort to enable a basic hybrid standard, such that online participants are granted some degree of active participation. This could include a video call platform, a text chat platform, a screen in the room showing online participants and cameras in the room showing both speakers and the room. It is worth noting that a key element in the above working well is active management by organizers of both audiences and by encouraging both in-person and online attendees to interact via the bridging platforms.

When it comes to why hybrid is thought to cost so much, this generally comes from the context of commercial conference venues and professional conference organizers (PCOs), especially in cases of parallel sessions and large numbers of attendees. But often the expensive solutions provided in these cases do not meet many of the basic needs of online attendees, and focus too much on the in-person experience alongside costly AV-heavy solutions. Fundamental questions that organizers need to ask in these cases is whether the conference format being

adopted is truly the right approach, or whether the approach could be innovated upon to better suit the needs of the community and if they are truly getting value for money for their quoted hybrid conference? Is a content-overloaded conference with numerous parallel streams really an effective way to bring an audience together, regardless of their mode of attendance? Are traditional PCOs in their current form really the right solution to handle the organization of meetings, given the dramatic impact technology has had globally on interaction? Are “TV-style” broadcasts and post-production effects really needed for academic talks, do we really need a full production crew for setting up a Zoom broadcast? Or is learning to do it ourselves combined with good lighting, a well placed camera, and an effective mic enough? One hope we have on the TFOM side is to see the rise of flexible approaches in hybrid that better meet the needs of our communities, but this requires organizers to see past the current offering of commercial hybrid approaches and seek solutions that truly help both online and in-person attendees get the most out of meetings.

Recommendations

- Consider the needs of both online and in-person attendees from the beginning of conference design and organization.
- Explore and experiment with low-cost alternatives (including technology) to commercial hybrid provision.
- Request solutions from PCOs that employ more innovative approaches to hybrid while clearly justifying the costs involved.

19.7.4 “The Energy Cost from Running an Online Meeting Is Just as High as an In-person One”

This simply isn’t true. When discussing the environmental impact of various meetings, this argument frequently arises as a criticism of remote formats, often without any more evidence than a hand-wavy sentiment that computers have carbon footprints too. While it is true that running personal computers, online event platforms, and remote data storage all come with an energy cost, these costs pale in comparison to those associated just with travel to and from conference venues, let alone all the other elements of in-person attendance including the production and waste of consumables (Tao et al. 2021). To put this in more concrete terms, for the online TFOM symposium hosted in 2020, the total estimated carbon budget for all 200 synchronous attendees combined was just 1420 kg CO₂e. This estimate aimed to be comprehensive, including pre-, during- and post-event estimates for all video processing, storage and streaming, device and server power usage, and even a fraction of attendees device manufacturing emissions. By comparison, this is less than the travel emissions alone of a single international attendee physically attending an in-person version of our event (Moss et al. 2020). This is worth repeating; the emissions for an entire 200 person online event were estimated as *less than half* the emissions of a *single one-way, economy-class flight* between Europe and Australia. Although meetings will always have an energy cost, online meetings for which the cost is dominated by electricity generation, can be made more sustainable by

ensuring that the electricity is generated from renewable resources. At present, there is no carbon-neutral way to fly, and for many electric vehicles are still out of reach (Burtscher et al. 2020, Tao et al. 2021). This misconception may be the result of awareness of the carbon footprint of *supercomputing*, which is indeed quite large.

Recommendations

- Research the carbon footprint associated with different formats for a given event, and make sure the cost is balanced by the benefits in format choice.
- Consider options for minimizing or offsetting the energy cost of an event.
- Include sustainability and carbon cost as part of the budget of a meeting or conference so that it is considered during every step of the process.

19.7.5 “We Need to Go Back to In-person Formats for the Sake of the Early Career Researchers (ECRs)”

Though often made with the best intentions, this argument is overly simplistic and falls into many of the same traps as section 19.7.2 above, and section 19.7.9 below. It presumes that future scientists will work in the same manner as their seniors, even though every profession continues to evolve. It also presumes that meaningful connections, which so often grow into career opportunities for ECRs, cannot be forged online. While there is unequivocal value in in-person meetings *for the right reasons*, blanket statements like this one are often made by senior academics whose views are informed by their own ECR experience in which in-person connection trumped all, but the views of ECRs themselves tend to be more nuanced and balanced (Köhler et al. 2022). To be sure, in-person activities are invaluable for ECRs as they work to establish a name for themselves in their chosen field, and these should continue to be made available to all (in moderation). However, we must ensure that any such in-person activities are both effective and widely accessible, and they must be supplemented by other forms of interaction to provide equitable opportunities to all ECRs, whether they can attend an in-person event or not. Just as importantly, we need to ask ourselves what it is about in-person events that is so powerful for ECRs, and why this hasn’t been emulated online; does it require different types of online interactions? A cultural shift amongst established academics in their attitudes to online events? Whatever the answer, taking steps to address this question will be integral to ensuring not only that ECRs survive in a digital-first world, but that they thrive.

Recommendations

- Make sure that changes to the meeting format doesn’t disadvantage ECRs
- Clearly identify whether your event will cater specifically to ECRs, and if it will, what elements will help them derive tangible benefits from attending.
- Consider ways to maximize the value of the in-person events that do happen for ECRs through structured networking and dedicated social time.
- Encourage senior academics attending all events (but especially online ones) to engage with ECRs beyond just delivering their talk.
- Encourage the academic community to move away from a model which leads ECRs to believe that in-person meetings are crucial to their career.

19.7.6 “COVID Showed Us How Tedious, Fatiguing and Boring Doing Everything Online Is, and How Ineffective It Is as a Format”

All of us can attest to having a bad experience with an online conference. Often this is because the format of the meeting was essentially the same as what would have been done in person. This is understandable because many conferences were quickly moved online when the pandemic started, not allowing time to think out format changes. What COVID actually showed us was that if every interaction becomes a video call, then we find meetings and interactions very tedious, fatiguing and boring. At the same time, the pandemic did dramatically accelerate the future of work because it showed, globally, that many professions could be carried out productively and effectively even with a distributed workforce. That said, interpersonal connections did suffer during the lockdown period because humanity has spent thousands of years meeting in person, and less than fifty years using computers to interact. It was also true that new starters faced challenges in particular because it was difficult for them to integrate into already established teams and earn trust when the bandwidth of interaction was more limited. However, none of these challenges do anything to show that online itself is ineffective as a format, but instead showed that how online formats were used was ineffective in some ways. It is not the case that everything should shift to online formats, but there are certainly varied and creative ways to use online formats, platforms and technologies effectively beyond just the video call.

Recommendations

- Schedule breaks between/during online meetings to reduce the impact of fatigue.
- Adopt more interactive formats and tools for meetings to avoid passive attendance.
- Make use of other formats (e.g. asynchronous chat) to reduce live meetings.
- Adapt meetings to be less structured and more informal where possible.
- Make time for social interactions in professional settings.

19.7.7 “We Don’t Need to Adapt, We Can Just Go Back to Normal Now because COVID Is over and We Don’t Need Online Formats in the Future”

What is “normal” is always tricky to identify when the world is continuously changing. It goes without saying that the pandemic has had a huge impact on the world far beyond its medical implications. For months, even years, a significant percentage of the working population was forced to be more distributed than ever before because of lockdowns and virus spread prevention. This dramatically accelerated “future of work” trends that had been observed and predicted for the years leading up. Diversity in both attendance and active participation was seen to increase for online meetings, and even now we have seen a majority of organizations and companies adopt hybrid working formats (if not embracing complete flexibility). This alongside the two equally important trends of mixed reality and artificial

intelligence point to a future of interaction that is both distributed and heavily dependent on technology. Not unlike the early 20th century declaration of Kelvin that the airplane would never be successful, the 1970s statement by Olsen that nobody would ever want a computer in their home or the pre-millennium prediction by Metcalfe that the internet would collapse before becoming useful, anyone who believes that COVID was a passing phase with no lasting impact and that we can go back to “normal” as if nothing changed seems disconnected from the reality of the trends across society. Even putting aside the context of the pandemic, a key point of this book is to prepare our communities for the rapidly rising impacts of climate change globally, and sustainability is a pivotal reason to accept the impetus to adapt that we have been offered. Combined with improving our accessibility and inclusivity as a global community, it seems obvious that online will be a critical part of our future. Those who adjust their ways of working sooner will be best placed in the years to come, especially with the changing economic factors that will equally demand changes in practice.

Recommendations

- Embrace hybrid ways of working as the future rather than a temporary situation.
- Provide flexibility in attendance as a default for all meetings, acknowledging that everyone’s needs are different.
- Take active steps to ensure that in cases of hybrid attendance, both the in-person and online audience are able to engage as equally as possible.

19.7.8 “There Are Too Many Technical Issues When You Go Online and They Often Can’t Be Solved”

While it is true that online meetings may run into technical issues unique to the medium, other meeting formats are not without their own pitfalls. It is not uncommon for small in-person meetings to be derailed by duplicate room bookings or an inadequate set up, and all meetings irrespective of format are susceptible to the dreaded AV problems. While we are quick to blame the technology if, say, a speaker cannot participate because their internet connection is down, we don’t apply the same criticism if a speaker misses the conference because of a canceled flight. As for any meeting, technical issues in an online meeting can be minimized with appropriate preparation. Having contingency plans can go a long way to ensuring a smooth running event, and online this can be as simple as preparing a backup meeting link and checking that key meeting attendees (such as keynote speakers, session chairs, etc) are familiar with the platform before the event starts. Remember good preparation is always apparent to an audience, and typically your audience will be more understanding of technical difficulties when they can see you have taken steps to mitigate them beforehand.

Recommendations

- Do not hold your online meetings to a higher standard than in-person ones, and remember things go wrong all the time, regardless of format.

- Spend adequate time safeguarding against predictable technical issues early as part of your planning stage. These include things like difficulty navigating the platform for attendees, and access issues related to meeting materials.
- Provide low-tech options for those hampered with technology issues e.g. live streaming sessions of platforms people have difficulty accessing.

19.7.9 “You Can’t Start New Projects or Form Effective Collaborations Online”

On the contrary, it is entirely possible to meet new collaborators and form effective collaborations online. However, just like with in-person events, the ease with which this can be done is heavily impacted by the attitude of both attendees and organizers. If no provisions are made for networking as part of an event, either in-person or online, then it will naturally be more difficult to meet new people than if there is a dedicated social space. Likewise, if attendees come to an event without any intention of engaging in networking or making introductions, then it will be very difficult for others to get to know them, regardless of the format. Since much of the recent shift to online was driven largely by the sudden emergence of the COVID-19 pandemic, many peoples’ first experience of organizing an online meeting was hasty and fuelled by a need to produce a minimum viable product. As a result, many recent online meetings have focused purely on content delivery at the expense of providing space for networking, which is taken almost for granted as an element of in-person events, whether it be in the form of a networking hour, or tea and biscuits following presentations. Admittedly networking online often requires a different type of interaction to what is expected in-person, but already many younger researchers are more adept at navigating these online pathways, suggesting their prevalence has only to grow.

Furthermore, a number of world-leading collaborations in astronomy are globally distributed: EHT, Planck, SKA, and the list goes on—and international collaborations are by no means limited to the astronomy domain. If it were not possible to collaborate effectively online then many of these large-scale collaborations would quite rapidly cease to function. In fact, astronomy and academia more broadly have a very long history of remote collaborations, with first private letters, then scientific journals providing means for connecting researchers across the globe. When viewed through this lens, making new acquaintances and collaborations via online channels is merely the next logical step in this long history of remote collaboration.

Recommendations

- Deliberately build time for collaborative networking into your events where appropriate.
- Bear in mind that online networking may require a different skill set to the in-person version.
- Be proactive in your approach to online networking much as you would in-person.
- Use tools that facilitate multiple ongoing conversations at the same time, with user controlled join and leave options for each conversation.

19.7.10 “The Technological Requirements Needed to Go Online Excludes Regions of the World Without Good Internet”

Similar to section 19.7.8, online meetings have a high reliance on technology for success and are lambasted more for technical issues. While limited in many parts of the world, it is *far* easier to obtain access to the internet than to find the resources for travel. Wifi is much cheaper than a jumbo jet. As well as having contingency plans for technical hiccups, for global audiences where technology may be an issue, alternative low-tech options can be provided. For example, for VR events there are platforms that offer 2D access for those who cannot afford or are uncomfortable with headsets. Similarly, one can choose content and delivery to minimize bandwidth demands. For platforms that have high computer demand, alternative lightweight access through livestreaming the session or screen sharing Zoom may also help. The participant may not have the same interactivity with the platform but at least they can feel some involvement. The issue is also slowly being mitigated, the global pandemic and the reliance on technology for everyday life has driven a shake-up of the technological infrastructure of many countries, and indeed nearly all countries now have technical access in urban areas to internet speeds in the low tens of megabits per second (World Population Review 2023). The issue is therefore more related to financial and rural access to those speeds, but this is a problem that can be solved far easier and cheaper than visas, using accessibility grants, and will often yield longer lasting impacts in situ than providing travel and accommodation costs. Many initiatives along these fronts already exist, and indeed even large scale upgrades are possible in this manner, with projects like the African SKA expected to result in significant upgrades to the technological infrastructure of much of the continent (Bhagal 2018).

Recommendations

- Provide low-tech options for those hampered with technology issues.
- Offer accessibility grants where possible to support people who have less access to technology (e.g. internet, hardware, software).
- Actively engage with communities to determine what their accessibility needs are when it comes to technical requirements of participation.

19.7.11 “There Are Some Things You Really Just Can’t Do Online and Have to Do in Person, Like Whiteboarding and Eating Together”

One of the challenges we regularly see in online interaction is that it is constantly being compared to, or trying to reproduce, the equivalent in-person interaction. In actuality, there are some activities that are better suited (currently, with contemporary technology) to in-person. Eating meals and drinking together is a good example of something that is much easier to do in person than, for example, in a VR headset. This may not stay true in the coming years if affordable and effective augmented reality options become widespread, since AR allows you to project virtual elements into a physical environment (and one could imagine, for example, eating a meal while sitting at a physical table, sitting next to a holographic projection of a far-away

friend or colleague). In the case of whiteboarding, we have actually found that online methods (either in browsers like Miro/Mural or in VR) can be much more inclusive than gathering around a physical whiteboard, depending on how they are used, although in some cases it is true that the in-person version is more effective. A major part of moving to more sustainable meeting formats is learning to decipher which activities are best suited to in-person gatherings, and which can be accomplished just as (if not more) effectively online. Once you can do this, you can make an informed decision about the best format to accomplish the goals of your next meeting. The conclusion here is that we should be using in-person formats for when they are most effective, and looking to online formats for where they can be equivalent or even superior—but we should also maintain an open mind, because technology is evolving rapidly and there are increasingly fewer examples of activities where “only in-person can be effective.”

Recommendations

- When choosing in-person formats, ensure these are activities that truly are most effective when carried out in person.
- Explore options for similar online formats to meet the same goals as the desired in-person activity.
- Regularly reassess the assumption that something best done in person has remained so in the face of advancing technology.

19.7.12 “You Can’t Solve Time Zones or the Fact that We Have a Round Planet, Therefore You Can’t Really Do Things Globally While Online”

This is a fundamental problem. For in-person meetings we experience it as jet lag. Who of us hasn’t nodded off during a session because their biological clock was still in a different time zone? But there are ways to address it. One of the advantages of the virtual format is that interactions do not need to happen in a live or synchronous manner. Many organizations in both academia and industry have maintained a strong global presence for decades thanks to judicious use of both synchronous and asynchronous meeting formats. In fact, there is no compelling reason why much of the ‘passive’ content seen frequently in live meetings like conferences (talks, presentations, etc) could not be delivered asynchronously, with a reduced but more efficient live meeting scheduled for the interactive discussion that typically follows these events. At present we rely heavily on live meetings for many kinds of information transfer, but moving away from this and to a more flexible form of work will only strengthen our ability to form and maintain global networks.

On a more practical note, in reality it is often the case that meeting participants will hail from some subset of all global time zones. If a short or one-off event does require synchronous interactions, finding a suitable meeting time is usually fairly straightforward, and can be done simply if attendees’ time zones are collected as part of the event registration (or during discussions prior to setting a time, for less formal interactions). For slightly more flexibility, it is worth considering collecting attendees’ *preferred working hours*, rather than just their timezone, as some individuals will be naturally inclined to working late at night, and others will have a preference to

start early. Obtaining this information makes it simple to accommodate attendees from a range of timezones with the help of a basic optimisation script (or a student happy to code one up for you). When an event has a truly global audience, it may also be useful to consider two repeat sessions separated by 8-12 hours, in order to allow all attendees the chance to participate.

Recommendations

- Schedule live meetings only for things that cannot be achieved asynchronously.
- Consider the actual time zones (or preferred working hours) of attendees when scheduling live meetings, which may include asking them directly via a survey.
- Very occasionally (and with enough notice), schedule a meeting which takes place as a single time to bring an entire community together (to avoid segregating regions), and rotate the timing in future to avoid locking in disadvantage.
- As an attendee, consider requesting funds for a local hotel or additional support for childcare, etc, even for a remote conference where you will not need to travel, especially if it is unsociable hours.

19.7.13 “There Are Too Many Tools Used Online and It’s Too Overwhelming and Inefficient”

There indeed are many tools used online and it can be overwhelming. This is in part because virtual connectivity tools are still in the innovation stage, with new ideas and techniques continuously being developed. But this problem also occurs with in-person meetings, as there are also many different, evolving ways of facilitating them as well. The existence of these many tools is driven by different goals and purposes, and as such organizers too should be selective in their choices. From the point of view of an online attendee, it is true that being confronted by many tools, especially unfamiliar ones, can be confusing, but a counterpoint to this is that a reluctance to learn or try new things goes against the spirit of scientific experimentation and limits agility and innovation. That said, even excessive use of familiar tools can require multiple windows and tabs to be opened, which can be inefficient and so it is good to minimize the use of tools. When introducing new tools to an event, organizers should make sure they are clearly explained, allow time for troubleshooting and have plans for when certain institutes or countries refuse to allow a tool to be used. This does not mean however that we should always stay with the lowest common denominator of toolkits, as IT policies are partly shaped by demand, and many new tools have not been encountered enough for a policy to be put in place. Your department may be happy to do so given enough time and demand. Finally, depending on the complexity of the tool, organizers should also offer “onboarding” or familiarization sessions.

Because of the challenges of mixing and matching tools for conducting an event, it may be tempting to use a single monolithic platform approach instead. But there are downsides to this approach as well, namely that no one platform is good at every

aspect. Some may provide excellent video streaming, others a better Q&A solution. So instead of overwhelming attendees with millions of tools, or stifling your event within the “walled-garden” of a single application, ensure instead that each additional tool used is backed by a clear, justifiable need and even more importantly, ensure that in all cases there is one (and preferably only one) tool for doing each thing, and that it is made obvious to the speakers, hosts and attendees.

Recommendations

- Carefully assess the need for a variety of tools and minimize the numbers necessary to meet the goals of a meeting or event.
- Organizers introducing new tools to their event should make sure they are clearly explained and that people are given time to gain familiarity.
- Balance the desire for an “all in one” platform against the need to use the right tool for a given approach, and experiment to find the best solution.

19.7.14 “Metrics for Career Progression Depend on Meeting In-person E.g. Number of Conference Presentations”

This is unfortunately probably quite true both directly, as in the case of conference presentations and indirectly through networking with the “right sort of people” to advance your career. Furthermore, in some disciplines conference proceedings are valued highly. Though reliance on in-person attendance for such conferences seems to be diminishing, there is still an inherent bias through favoring in-person presenters for accepting talks or high virtual registration fees ostensibly to subsidize in-person costs. Also, the number of publications arising from a conference has been cited as a measure of success of that conference and related to the various concerns about the importance of networking. Across academia there have been many discussions about diversifying the metrics for success to be more inclusive of the full contributions made by a given academic, and it is clear that this aspect falls under that umbrella of needing widespread improvement.

Recommendations

- There needs to be a continued overhaul of how academic performance is measured to something more equitable. ECRs should not be encouraged or required to travel to be successful.
- Conference proceedings, presentations and contributions delivered virtually should carry equal weight to those taking place in person.
- Build more global impactful networking initiatives that do not rely on in-person attendance.

19.7.15 “Online Would Be Better and More Effective If It Felt More Like Being in Person”

Early in the TFOM process, we were introduced to a famous human-computer interaction paper from 1992 titled “Beyond Being There” (Hollan & Stornetta 1992), in which many of the challenges we felt were unique to the pandemic era of 2020 were identified back in the early 1990s. In this paper, it was pointed out that one of

the biggest impediments that we arbitrarily impose on online formats is the requirement to reproduce in-person interaction, as if that is the end goal that should determine success. However, one of the most prevalent and impactful elements of online interaction is semi-asynchronous text communication (e.g. being able to send instantaneous messages which someone may, or may not, reply to right away). This is an example of something completely impossible to do “in person,” and something that really changes the way we interact with others both locally and globally. Based on this, we have adopted this sentiment in much of our work with online formats, constantly revisiting the assumption of online as a reproduction of in-person and asking ourselves: “How do we go *beyond being there* with online formats?” Online is not about just reproducing what we would do in person, but about opening up possibilities to do *more* than we could ever achieve with in-person only interaction.

Recommendations

- Consider what is possible online that might not be possible in person and make use of that as an advantage of the format.
- Ensure that the format choice for a meeting is driven by the goals of the meeting rather than by the established norms around how things typically take place.
- Adopt a blend of different formats to meet the different needs of a community (while constraining choices in the context of accessibility, inclusivity and sustainability).

19.7.16 “The Online COVID Version of the Conference Didn’t Feature That Many New Diverse People, So It’s Not True That Online Actually Improves the Diversity”

Actually, it is widely reported in the literature that diversity was measurably increased for many conferences and meetings that shifted online during the height of the pandemic (Skiles et al. 2022). The removal of traditional barriers of access meant that conferences which typically excluded many were suddenly available, particularly for marginalized groups. In many cases where the statistics of a meeting were analyzed, diversity factors such as gender, seniority, country of origin and country of residences changed significantly for the positive during the pandemic. In cases where the statistics did not, it is important to recognise that a traditionally-excluded community does not necessarily become instantly included just because of the door being open. In this sense, the difference between something being accessible and being inclusive is critical. Diversity starts with accessibility (e.g. ensuring that your meeting choices are such that everyone *can* take part), but it is only effectively achieved with inclusivity (e.g. the meeting is structured in such a way that diverse people feel empowered *to* take part).

Recommendations

- Define systematic and structured ways to measure and analyze meeting attendance (including consideration of ethics), such that diversity of participation can be tracked over time.

- Always include online attendance as a default to anything that takes place in person, and aim for online attendance which enables people to participate as much as possible.
- Actively support and engage under-represented people to ensure that they feel welcome and included to take part in the relevant meetings and conferences.

19.7.17 “Opportunities for Research and Collaboration Often Appear during Informal Networking Over Drinks or in Other Social Settings, So These Opportunities Won’t Be Available in Online Meeting Formats”

It is true that, given thousands of years of interaction taking place in person, many people are generally pretty good at in-person socializing compared with the online equivalent. We are also more inclined to build relationships and trust people quicker with certain cues that are easier achieved in person (Kramer 2009, Kalia 2021). So given this, informal networking does play an important part in establishing new collaborations and forming relationships, and traditionally this has been well-achieved via in-person social settings. However, there are a few caveats to consider here. Firstly, most of the evidence for this is highly anecdotal, and has not been measured well either qualitatively or quantitatively for either in-person or online formats, and this is something that should be amended in the coming years. Secondly, this kind of statement does not acknowledge the dark side of relying on these kind of interactions to gain opportunities—are we saying that, as a community, we are happy with the idea that those who cannot take part in person (for all the reasons we know about when it comes to accessibility and inclusivity, including cultural differences around acceptable food and drinks) do not deserve the same kind of opportunities? And lastly, the world and opportunities within it are immense, and there are so many important and interesting challenges to solve together. It seems a great missed opportunity in itself if we choose to limit ourselves only to the opportunities that arise when we can physically gather in one location.

Recommendations

- Deliberately and considerately plan activities and events which aim to provide these kind of informal opportunities to people online as well as in-person
- When organizing in-person social events, consider groups and minorities who might be traditionally unable to take part and support their inclusion where possible
- Explore possibilities both online and in-person to provide ways in which these kind of opportunities may be realized in different or more direct ways (e.g. team-building, careers fair, social activities, games, etc)

19.8 Case Studies in Online and Hybrid Design

We have found in our activities that often the lack of widespread examples in how to approach online or hybrid interaction creates barriers for people designing meetings. In this section, we share case studies of some highlight events we have either orchestrated or been involved in, with the goal of sharing the context behind it, the

goals, the unique elements of a particular case study and also the aspects which worked well or were challenges.

The case studies below were selected to cover a range of different scenarios, and we aim that they provide a useful insight into some of the broad possibilities in online and hybrid interaction, but we note that they are a small subset of the wide variety of creative and innovative approaches that have been documented. We encourage the reader to look beyond these case studies and seek further inspiration in the literature and online when it comes to the current state of best practice for online and hybrid meetings.

19.8.1 The Future of Meetings Symposium (TFOM, 2020)

“The Future of Meetings” symposium was held over three days (15th–17th September 2020) as a CSIRO Cutting Edge Science Symposium (CESS), originating from pre-pandemic discussions around sustainability and accessibility amongst astronomers and motivated by a desire to collaborate more effectively virtually (Figure 19.3). During the preparation of the proposal, COVID-19 happened and thus made the topic suddenly relevant to a much broader range of people, strengthening themes of the future of work and distributed collaboration. The lack of a precedent combined with being the first online-only CESS meant planning and design started from scratch, which fostered a culture of experimentation and innovation.

After considerable research into available technology, the [Whova](#) platform was chosen as a conference platform on the basis of features, cost and flexibility. Whova provided many desirable elements such as attendee management, timetabling including embedding recordings, streaming and Zoom meetings into the schedule, and extensive community features for keeping attendees connected. Most



Figure 19.3. The “expression of interest” page for the TFOM symposium in 2020. The current community of practice was formed by a subset of the organizing committee who stayed active after the symposium to continue to advocate for and experiment with new meeting technologies.

importantly, Whova as a platform gave quite a bit of control and autonomy to organizers, which was extremely important for oversight and management of the event. While the platform backend is slightly complicated, it includes many elements of automation to streamline workflows and provides an effective comprehensive solution that was well-suited to TFOM.

The three days were themed “Today, Tomorrow, Future” and made up of a combination of invited talks, panel discussions, workshops and social sessions. The majority of talks were presented as live-streamed recordings with the speakers available online for live Q&A via the Whova platform. Other live sessions such as panel discussions, extended live Q&A and social sessions were held through Zoom and Gather. Legacy recorded content can be found at the [TFOM](#) archive and lessons learned from planning and delivering the symposium have been made available in a publicly available report (Moss et al. [2020](#)).

What worked well: This was aimed to be an innovative symposium in design and execution, and gained a wide audience (over 1100 attendees). It showed the possibility of making virtual conferences effective. The diversity of the audience introduced perspectives from beyond the normal circle of attendees giving rise to interesting and engaging discussions.

Limitations: This symposium was held in the early days of the pandemic and so there was a general lack of familiarity with online tools and format. Similarly, unfamiliarity meant there was not as much participation in the social sessions as hoped from people literally “switching off” after the formal content of the day.

19.8.2 International Conference on Women in Physics (ICWIP, 2021)

This was a fully virtual international conference, held 11–16 July 2021, but unlike the TFOM Symposium was part of a series that had previously been held only in-person and, furthermore, was designed to be accessible to participants from developing nations. The timetable for the conference was designed with a global audience in mind so that no specific time zone would be disadvantaged and the impact of time zones mitigated by making all content available for asynchronous viewing. The Whova platform was again chosen for this event and the community pages proved particularly popular, keeping attendees connected through sharing photos and discussion. The social aspect, which was traditionally a key part of the conference, was facilitated through a variety of social sessions from indigenous dance lessons in Zoom to trivia nights and Penguin Parade watch parties in Gather. These activities had been planned for the original in-person conference before it was postponed by the COVID-19 pandemic, and the participants appreciated that they could still have some form of local Australian experience. The Gather space was available throughout the conference and many of the participants made use of it for their personal social and work-related meet-ups.

What worked well: The social aspect where the community enthusiastically engaged in the organized activities and furthermore arranged their own. The well-attended poster sessions were by far one of the best demonstrations of the power of spatial-interaction platforms like Gather (Figure [19.4](#)), with clusters of people

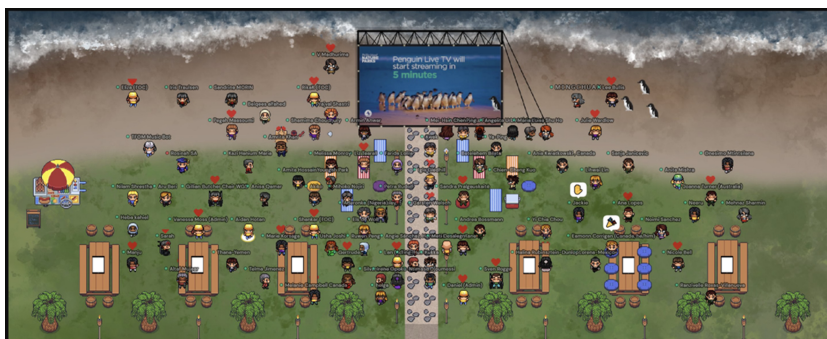


Figure 19.4. Attendees of the ICWIP 2021 conference in Gather, assembled for the conference photo.

scattered throughout the virtual poster room having conversations at the designated poster session times.

Limitations: ICWIP is traditionally aimed at developing nations so there were some issues with technology and familiarity with online platforms. As a conference series, there were many stakeholders involved in the structure and format which resulted in some constraints when it came to the approach taken.

19.8.3 Science Pathways 2021 (SP21, 2021)

“Science Pathways” is a national-level conference run by the Early and Mid Career Researcher (EMCR) Forum, which is a representative body that exists as part of the Australian Academy of Science. It is one of the only meetings that seeks to unite junior STEM researchers across Australia, and traditionally takes place every two years. The 2020 meeting was postponed to 2021 due to COVID, and the planning for the 2021 meeting took place while the pandemic was still very much active in Australia. The executive determined that SP2021 should take place in a mix of formats to meet the different needs of EMCRs, including hybrid, in-person only and online-only. As such, a hybrid launch half-day was designed (taking place in Melbourne in late November 2021), followed by a day of in-person workshops, and then in early December there were three online-only half days featuring panels, workshops and social activities. The hybrid launch day was designed to minimize the gap between in-person and online audiences where possible, such as having all questions asked via the conference platform only and ensuring all attendees were able to access and use the online platform. In addition, a member of the organizers was designated as the in-person representative of online attendees, and this member attended online while sitting in the room adjacent to the main conference venue (Figure 19.5). This meant that if any issues arose for the online audience, there was someone local aware and ready to take action immediately. For those in person, a visit to a local museum “behind the scenes” was arranged, and the workshops the next day were chosen with the goal of leveraging the in-person nature of attendees. The online-only days were scheduled to be a little spread out in time, meaning that there was the building of an online community between the launch day and the



Figure 19.5. To help bridge the gap between in-person and online attendees during the hybrid launch day, one member of the organising team attended “online” from the venue, acting as the in-person representative of the online audience.

online days, with discussions and interactions continuing beyond the “live” time of the conference via the platform.

What worked well: The mixed mode of formats meant that there were elements of the meeting that appealed to everyone. Having the conference spread out in time kept the community together, facilitated by the technical platform, which stayed active before and after the meeting and allowed people to interact and connect. The hybrid design, which aimed to minimize the degree to which in-person attendees were favored, meant that both audiences were active participants during the launch day.

Limitations: The in-person elements were made possible by generous sponsorship from the host institution, covering venue, catering and AV, which otherwise would have been extremely expensive compared to the other costs associated with running the conference. As with other online meetings, getting attendee buy-in for the online social activities was difficult and there was low turn-out in general for these, even though many people expressed desire for informal unstructured interaction online.

19.8.4 Infrared 2022 (IR2022, 2022)

“An Infrared Bright Future for Ground-based IR Observatories in the Era of JWST” (IR2022) conference was hosted by the JAXA Institute of Space and Astronautical Science and held online between February 14th–18th 2022. This was the first astronomy conference that the TFOM community partnered with,



Figure 19.6. TFOM worked with IR2022 organisers to build a VR-compatible add-on venue in Altspace, which was used to host posters, talks, livestreams of the sessions and a JAXA exhibition display.

using the gained experience in virtual reality to provide a social venue and exhibition space alongside the planned conference program which took part primarily in Zoom and Slack.

The conference organizers were keen to provide an area where attendees could interact, both as part of the conference poster session, and also for general discussion and networking. A 3D virtual space with multiple rooms was set-up using the Microsoft AltspaceVR platform (Figure 19.6), which provides a variety of ready-made conference environments that can be freely used. The venue could be accessed via a computer or VR headset, with visitors creating an avatar that allowed them to move around the area. As this was a 3D space, directional audio and audio fall-off allowed simultaneous overlapping conversations in the same room, as with an in-person venue.

Two rooms in the virtual venue were dedicated to posters, which were displayed as a large size and included embedded videos of the poster presenter summarizing their results. Another room included a stage where two live (in platform) talks were held, and the live research talks from the main zoom stream were also displayed for attendees who wanted to watch the presentations together. A fourth room allowed visitors to watch the pre-recorded talks together on-demand to encourage group watch parties, re-watching of key talk sections, and ongoing discussions. The final room was designed as an exhibit space for space missions from JAXA. This included videos, images and 3D models of spacecraft which could be handled by visitors. The exhibition space was particularly popular for networking, as attendees could explore the space and fall into conversation with others also examining the same objects or media.

While Zoom was still the main platform for the conference's presentations, the virtual space offered a natural venue to engage with other meeting participants, and

the exhibition space was an attraction that could not have been easily implemented for an in-person meeting.

What worked well: The exhibition space was particularly successful at encouraging people to attend the virtual socials, and the models and exhibits were good ice breakers for seeding conversation.

Limitations: A few participants were not able to access the virtual space from their computer, due to struggles with installing the software. This could have been solved by stronger encouragement to install the software in advance, or adopting a platform that was browser-based.

19.8.5 Science for Public Good x TFOM (S4PGxTFOM, 2022)

Throughout 2022, TFOM hosted a series of events centred around the theme of “Science for the Public Good” funded by a grant from the Theo Murphy Initiative. As part of this series, four events were held solely in a virtual space built in Altspace, a VR platform compatible with both headsets and desktops (which has since been unfortunately retired). These events featured speakers drawn from a wide range of scientific disciplines, and touched on some of the many ways in which different areas of science, from astronomy to ecology, can build towards a greater public good. The events were held live and scheduled so that the series spanned all time zones, with at least one event falling during business hours for attendees from Oceania, Europe and the Americas, and it was intended from the outset that social connections and networking would be integrated into each event. This desire for social connection was in part what led to Altspace as the venue of choice, because unlike a simpler video conferencing platform, the 3D virtual venue could be easily transformed from presentation space to networking space. What is more, the spatial audio element allowed for organic conversation groups to form after the talks were concluded, with additional immersion possible for those accessing the events via VR headset thanks to cross-platform support.

Additionally, a fifth, in-person event was held in conjunction with the 8th Australia-Spain Research Forum “A Digital World” to introduce in-person attendees at the workshop to the possibilities of meeting and collaborating in VR spaces (Figure 19.7). Five VR headsets were set up in a room at the workshop venue and all connected to the same Altspace event venue, alongside a PC connected to the same venue and projected on a wall. This allowed dozens of conference attendees to explore the VR event space during breaks in the in-person workshop.

What worked well: The platform was easily accessible both in VR and via desktop, and the venue seamlessly transformed between presentation and socialization space. Coordinating an in-person event led to much greater VR exposure, and feedback on the VR immersive experience and ability to more naturally interact/socialize was very positive.

Limitations: The platform was not accessible on low-spec systems or sometimes blocked by institutional firewalls, creating a barrier for access. Although the series spanned time zones, events were synchronous so not all events could be attended by



Figure 19.7. During 2022, a series of events were run in VR as part of the “Science for Public Good x TFOM” collaboration, where attendees could join via virtual reality, desktop or livestream.

all attendees. Visitors from the in-person workshop only had a superficial introduction to VR events due to time constraints.

19.8.6 Astronomers for Planet Earth (A4E, 2022)

The Astronomers for Planet Earth held their inaugural symposium fully virtually on 28 Nov–2 Dec 2022 (Wagner et al. 2023). The program was designed to be inclusive of all time zones by allowing asynchronous access to recorded talks and offering two daily live discussion sessions in Gather (Figure 19.8). The opening and closing sessions, and a panel discussion with invited representatives from large astronomy and space consortia, were broadcast live via Zoom and YouTube, but recorded for those who couldn’t attend. Furthermore, members of the A4E community were encouraged to organize independent workshops of which four were held. All recordings from the symposium are available on the A4E [YouTube](#) channel. The event was judged a success with live audiences of as many as 150 people and over 4,000 views of the recorded material. The level of engagement over the duration of the event showed a clear interest of people in sustainability and the online symposium allowed them to share their views with a wider like-minded community. This particular point was commented on by many who had been feeling increasingly isolated for holding such concerns.

What worked well: The discussion sessions were well-attended and generated lively discussion from a global audience. The “pre-watch” format with focus on discussion allowed for the consumption of the content at an attendee’s own pace as well as some introspection before the open discussion.

Limitations: People still do not dedicate the same time and energy to online conferences as they would in-person events (especially social events and activities), and the global time zone scheduling inevitably inconvenience some.



Figure 19.8. The A4E symposium aimed to maximise interaction time during the live sessions, with parallel discussion sessions taking place on a variety of topics.

19.9 Looking to the Future

As we have discussed above, the modern era has seen vast improvements to communications technology that have fundamentally changed how we work, live and socialize, making it easier than ever to interact across long distances without the monetary and climate costs of moving people. Yet despite these improvements we still travel for work far more than we should, with travel for conferences and meetings being two of the biggest drivers of academic carbon emissions to date.

Reducing these emissions should be a major priority. Many people still feel that in-person meetings bring a vital human connection, a sense of presence to collaborations that is just not reproducible remotely. This tension is an issue. While a return to purely in-person events would allow for that important in-person networking, it will also exclude many who are unable to travel, while driving up our current (post-pandemic) carbon emissions by several orders of magnitude. The solution then, is not to give up on remote collaboration. Instead we must find a way to bring the best parts of in-person meetings to our remote ones.

Virtual collaboration tools are very much in the “innovation stage,” and like many of the technologies we take for granted now, e.g., CCD detectors and digital projectors, these tools are not yet fully ready—often frustratingly so. Luckily, communications technology has not stopped with the development of the video call, and in the decades since the video call’s popularization (let alone creation) there are now many more advanced solutions available for remote work and social. The most promising of these, Extended Reality (XR), brings huge improvements in interaction quality, particularly in making users feel connected and making them feel like they



Figure 19.9. ¹ British inventor John Logie Baird and his first publicly demonstrated television system, with which he transmitted moving pictures in 1925. In a few years is this what the early days of virtual conferencing will seem like?

are really ‘there,’ while also opening up opportunities to go beyond what is possible in person.

Virtual collaboration *formats* are also in this stage of innovation. There’s no reason why we need to do conferences online as we do them in person—we’re just doing them that way because that’s what we know. A good analogy is the early days of television. When it was invented they didn’t know how to do TV shows, but they did know how to do radio shows. So they did radio shows on TV. Over time people eventually recognized the power of the new technology, and new show formats developed. That’s where we are now with virtual conferences (Figure 19.9). So keep that in mind if you’ve had a bad experience online.

It’s exciting to think about what the future can bring. XR, encompassing both Virtual Reality (VR) and Augmented Reality (AR), is leaps and bounds ahead of what most people think, and the impact it will have on our world as it develops will be substantial. This is because at its core XR is not a game or a social platform, it is the next ubiquitous computing platform, subsuming laptops and smartphones to provide not only the functionality of both, but combining them with true spatial awareness and presence (e.g., Figure 19.10).

It is hard to overstate just how much our society might change when we can cast not just gigabytes of data around the world on a whim, but cast real-time digital

¹ https://en.wikipedia.org/wiki/John_Logie_Baird#/media/File:John_Logie_Baird_and_Stooky_Bill.png.



Figure 19.10. Interacting with digital versions of physical hardware (such as telescope components) is already easy to do in existing platforms like Spatial.

representations of both *ourselves and our surroundings* anywhere, to meet anyone, using nothing but a small headset and an internet connection.

Imagine the ramifications of being able to keep access to all of our current screen based technology while also enabling the feeling of true human connection that is so missed when staring through a screen. How will our world change, when we are able to walk around campus with a photoreal, 3D, real-time view of a colleague half the world away walking alongside us? What will happen when you can have a virtual meeting with eye contact, facial expressions, body language and spatial audio in a shared environment that seems real?

The applications of this in astronomy and beyond are vast. You could join a new colleague for dinner remotely (Figure 19.11), meet a potential collaborator for a round of virtual mini-golf (while discussing the latest paper of course), explore a 3D scan of an asteroid by standing on it together, whisper a question to a mentor while listening to a talk in a virtual lecture theater, or mingle with new and interesting people while exploring a virtual expo hall. Once this is possible, will we still regularly fly halfway around the world to sit in an office or lecture theater? Would our funding departments still approve travel?

The thing is, most of this already *is* possible with current XR headsets, and while it is yet to be rolled into ‘one headset to rule them all’, it is getting close. Current headsets are still forced to make the trade-off between graphical quality and form factor, convenience and cheapness, but this will not stay the case for long. Already one can have a decent fully-functioning standalone headset for a few hundred USD, or a spectacular headset with a supporting PC for a few thousand. Daunting though these numbers may seem, both are one-off costs well below those associated with attending a single international conference.



Figure 19.11. Virtual reality provides a new mechanism for organic distributed social interaction that has the potential to go well beyond what is possible in person, such as this example from “Walkabout Mini Golf.” The immensely popular app offers convincing mini-golf physics and multi-user interaction in exotic and beautifully constructed worlds.

Indeed, with current headsets it is already possible to have a meeting, even a conference completely in XR (e.g., Figure 19.12). To have a true virtual stage with speakers, laser pointers, powerpoints and microphones, to have attendees stand up in the crowd to ask questions leading to a natural back and forth, and to have posters and pre-recorded on-demand video playback. Most importantly perhaps, the sheer sense of presence and spatial audio provided by being in XR enables that vital component of organic networking and social that is so often missing in current online conferences; that feeling of being together.

But why should we settle for mundanely reproducing an in-person conference? An XR conference can give every attendee a front row seat, or let the speaker speak in one language and be heard in the attendees’ native tongue. It can provide conference badges that contain recent publications, interests and affiliations and provide posters that can be made interactive or even contain 3D visualizations, while still allowing the owner to stand in front of it to answer questions.

From an organizational perspective a XR lecture theater can be expanded or contracted on the fly to handle more guests or engender a more relaxed atmosphere



Figure 19.12. It is already possible using widely available platforms to have people gather in 3D immersive spaces, such as this space in Glue, where they can discuss, share data and collaborate on whiteboards together across VR headset and desktop devices.

(Figure 19.13), and in terms of accessibility the options for XR meetings (and more traditional online meetings) are far ahead of what is possible in-person. Whether it's something minor, such as allowing a speaker to reduce the apparent number of the attendees to deal with stage fright, to allowing vision impaired users to have built-in audio readers for the entire shared reality, XR meetings give us the opportunity to go beyond being there in supporting both attendees and speakers. Beyond all of this, however, simply by not requiring people to be away from their homes and duties for extended periods of time, XR meetings (like all online meetings) are just so much more accessible and inclusive than their in person counterparts, while still providing that all important “human” connection that we want in person for, at just a fraction of the financial and environmental costs.

So if this is all already possible, why has it not been done? The truth is it has, and *is* being done. There are many examples of effective conferences and meetings that are carried out in online VR or AR spaces by a wide range of academic, industry and government and social organizations. But the biggest issue with XR spaces is that most people's experience with them is via a backwards compatibility option (like a laptop or phone), and much like a video call where 90% of the attendees are dialed in by landline, the benefits of the better technology are not apparent unless the majority of users are connected using the new approach. The technology is also new to many, causing its own learning curve issues and triggering risk aversion strategies when doing larger events. In short, while XR conferences can solve many of the issues we have with online meetings, they unfortunately (much like video call conferences and meetings before the pandemic) currently suffer from a critical mass problem.

So at last we return to that all-important question: How can we help change our institutes, our governments and ourselves to help address climate change?



Figure 19.13. Members of TFOM gathered in virtual reality via headset and desktop to watch content together, despite being distributed across vast distances.

As astronomers we have a rather unique place as members of our truly worldwide research field. United by shared skies and international facilities, astronomy transcends borders and cultures in a way that just isn't that common in many academic or industry fields, and this means we are uniquely placed to push for better ways of doing things.

So we urge you, our readers, to push. Push for a better future. Advocate for your institutes to swap to renewable power supplies. Campaign for the time and resources to make effective use of more sustainable travel options, and speak out against inadequate organizational and political stances on climate, at all scales.

But add to your list making remote work, remote conferences and remote social events an integral part of your workplace. Not *just* because it is cheaper, but because every event you attend digitally is a plane not taken, a commute not driven, and a fellow scientist not excluded.

At the same time, acknowledge the current limitations of remote teams, and advocate for online meetings that make us feel more connected to each other. Use XR for what it's meant for; to spend time with all those you wish you could see more often, to share new experiences together, and to feel physically closer to your friends, family and colleagues, regardless of the miles and oceans between you.

Because by doing so, we have the potential to open the door to a better world. One brimming with new colleagues you might never otherwise have had the chance to meet, and old friends you might never normally have seen again. A world where remote conferences are as fun and productive as in-person, and where opportunities for scientists to share their work are available regardless of where they are born. A world that even with all of these improvements, is also hundreds of times more sustainable than our old.

So regardless of why you first picked up this book, whether you were looking for guidance on the state of our planet, seeking new colleagues who share your concerns,

or simply wanting something that you can do personally to improve our future world, and our time in it. We urge you; strive for better remote collaboration. Not only because it *will* help us do better science, but because we hope that by embracing these fantastic new digital worlds, we might just get to keep our real one.

References

- Aycock, L. M., et al. 2019, *Phys. Rev. Phys. Educ. Res.*, **15**, 010121
- Bhagal, N. 2018, How the SKA telescope is boosting South Africa's knowledge economy. The Conversation. <http://theconversation.com/how-the-ska-telescope-is-boosting-south-africas-knowledge-economy-96228>
- Burtscher, L., et al. 2020, *NatAs*, **4**, 823
- Carbon Independent 2007, *Aviation Emissions*, viewed 7 August 2023 at <https://www.carbon-independent.org/22.html>
- Curry, D. 2023a, Discord Revenue and Usage Statistics (2023). <https://www.businessofapps.com/data/discord-statistics/>
- Curry, D. 2023b, Microsoft Teams Revenue and Usage Statistics (2023). <https://www.businessofapps.com/data/microsoft-teams-statistics/>
- Duncan, N. W., & Shean, R. 2023, Analysing the effectiveness of Twitter as an equitable community communication tool for international conferences PeerJ **11** e15270
- Gokus, A., et al. 2023, submitted to PNAS Nexus
- Hollan, J., & Stornetta, S. 1992, CHI '92: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems 119
- IEA 2023, The world's top 1% of emitters produce over 1000 times more CO2 than the bottom 1% (Paris: IEA) <https://www.iea.org/commentaries/the-world-s-top-1-of-emitters-produce-over-1000-times-more-co2-than-the-bottom-1>
- Kalia, S. 2021, *The Swaddle* Available at: <https://theswaddle.com/why-we-trust-the-people-we-trust/>
- Klöwer, M., Hopkins, D., Allen, M., & Higham, J. 2020, *Natur*, **583**, 356
- Köhler, J. K., et al. 2022, *FrPs*, **13**, 906108
- Kramer, R. M. 2009, *Harvard Business Review* <https://hbr.org/2009/06/rethinking-trust>
- Moss, V. A., et al. 2020, *The Future of Meetings: Outcomes and Recommendations*, Zenodo (v1.0) <https://zenodo.org/record/4345562>
- NASEM 2021, *Engineering, and Medicine* (Washington, DC: The National Academies Press)
- Rommel, A. 2021, *Natur*, **591**, 185
- Skiles, M., et al. 2022, *Nat. Sustain.*, **5**, 149
- Tang, J. 2021, *Proceedings of the ACM on Human-Computer Interaction*, **5**, (CSCW1) 1
- Tao, Y., Steckel, D., Klemeš, J. J., & You, F. 2021, *NatCo*, **12**, 7324
- Vargas, S., et al. 2022, Astronomy for Accessibility and Inclusion Revista Mexicana de Astronomía y Astrofísica. Serie de Conferencias Vol. 54, (Universidad Nacional Autónoma de México)
- Wagner, S. M., et al. 2023, *NatAs*, **7**, 244
- World Population Review 2023, Internet speeds by country. <https://worldpopulationreview.com/country-rankings/internet-speeds-by-country> (accessed 15 August 2023)